

Sample TACACS+ Configuration for a Cisco Router

College Lab Setup

1. Basic TACACS+ Configuration

```
Router(config)# aaa new-model
```

```
Router(config)# tacacs server TACACS-SERVER
```

```
Router(config-server-tacacs)# address ipv4 192.168.10.10
```

```
Router(config-server-tacacs)# key cisco123
```

```
Router(config)# aaa authentication login default group tacacs+ local
```

```
Router(config)# aaa authorization exec default group tacacs+ local
```

```
Router(config)# username labadmin privilege 15 secret lab123
```

```
Router(config)# line vty 0 4
```

```
Router(config-line)# login authentication default
```

```
Router(config-line)# transport input ssh
```

```
Router(config-line)# exit
```

2. What Each Important Command Does

Command	Purpose
aaa new-model	Enables AAA (Authentication, Authorization and Accounting).
tacacs server TACACS-SERVER	Creates the TACACS+ server configuration.
address ipv4 192.168.10.10	Specifies the TACACS+ server IP address.
key cisco123	Sets the shared secret between the router and TACACS+ server.
aaa authentication login ...	Uses TACACS+ for user login authentication.
aaa authorization exec ...	Controls authorization for starting an EXEC session.
username labadmin ...	Provides a local backup account.
login authentication default	Applies the AAA authentication method to VTY login.
transport input ssh	Allows SSH instead of insecure Telnet.

3. Commands I Would Modify for a College Lab

A. Modify the TACACS+ Server IP

The example uses 192.168.10.10. In a Packet Tracer or college lab, change this to the actual IP address of the TACACS+ server in the topology.

Example: address ipv4 192.168.1.100

B. Modify the Shared Secret

The example uses cisco123. This is acceptable for a simple lab demonstration, but use the shared secret required by the lab. The same secret must be configured on the TACACS+ server.

Example: key MyLabTACACS123

C. Keep the Local Fallback

Keep 'group tacacs+ local'. If the TACACS+ server is unavailable, the router can try the local username database.

```
aaa authentication login default group tacacs+ local
```

D. Use SSH Instead of Telnet

Use SSH for secure remote management rather than Telnet.

```
transport input ssh
```

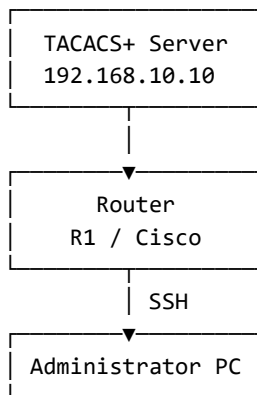
4. Older TACACS+ Command Style

Some tutorials show the older/legacy configuration syntax:

```
tacacs-server host 192.168.10.10  
tacacs-server key cisco123
```

Newer Cisco IOS configurations commonly use the named server style shown in the main configuration. For a college lab, check the IOS version and the available command options with '?' before choosing the syntax.

5. Simple Lab Topology



For a college lab, mainly modify the TACACS+ server IP address, shared secret, and local username. Keep aaa new-model, TACACS+ authentication, local fallback, and SSH. Packet Tracer's TACACS+/AAA support can vary by device and version, so if a command is rejected, check the available '?' options on the router.